

Intelligent Document Processing System

Extract text, tables & images from any document – automatically



FastAPI



Streamlit



Tesseract OCR



OpenCV



PyMuPDF



pdfplumber

The Problem

Manual document handling is slow, error-prone, and completely unscalable for modern workflows.

Unstructured Documents

PDFs, scans, and images each require manual reading and tedious data re-entry.

Time-Consuming

Hours spent extracting tables and text that a machine could handle in seconds.

Error-Prone

Human transcription introduces costly mistakes in critical financial and legal data.

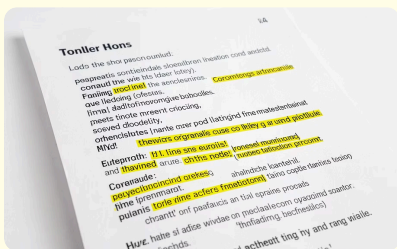
Not Searchable

Scanned PDFs are image-only – impossible to copy, search, or index programmatically.



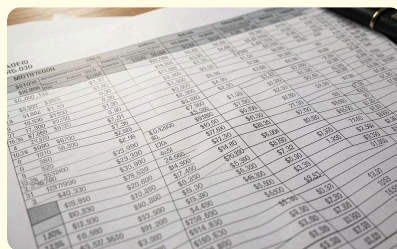
Our Solution

One upload. Automatic extraction of everything – text, tables, and images – from any document type.



Extracted Text

Searchable TXT + PDF output. Handles both digital and OCR'd scanned documents.



Tables → Excel

Auto-detected grid structure exported to multi-sheet XLSX files instantly.



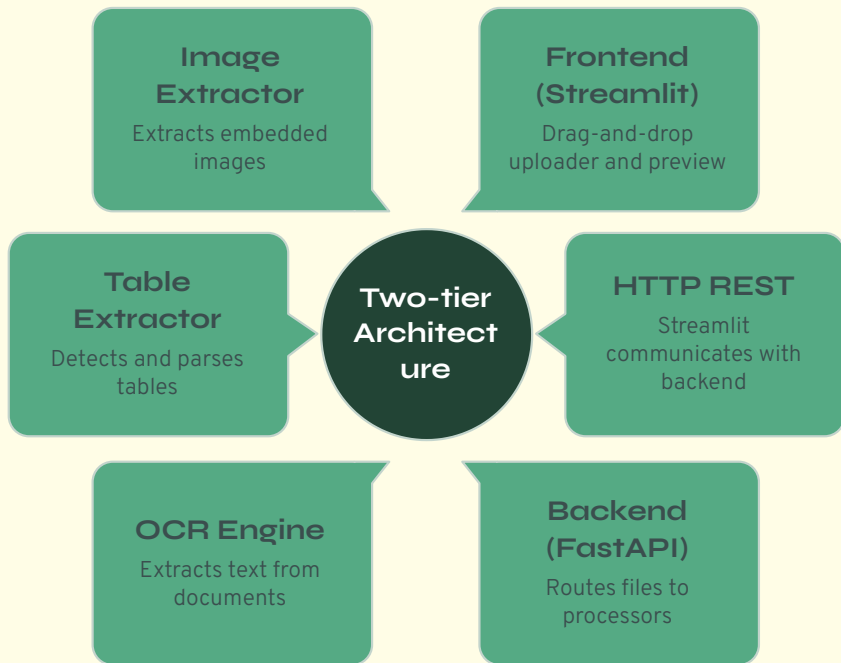
Images & Logos

Embedded images, logos, and stamps saved as PNG/JPG and viewable in the dashboard.



Structured JSON output · Downloadable from dashboard · Works on scanned & digital documents

System Architecture



Frontend – Streamlit

Drag-and-drop uploader · OCR preview · Table grid · Image viewer ·
One-click downloads

Backend – FastAPI

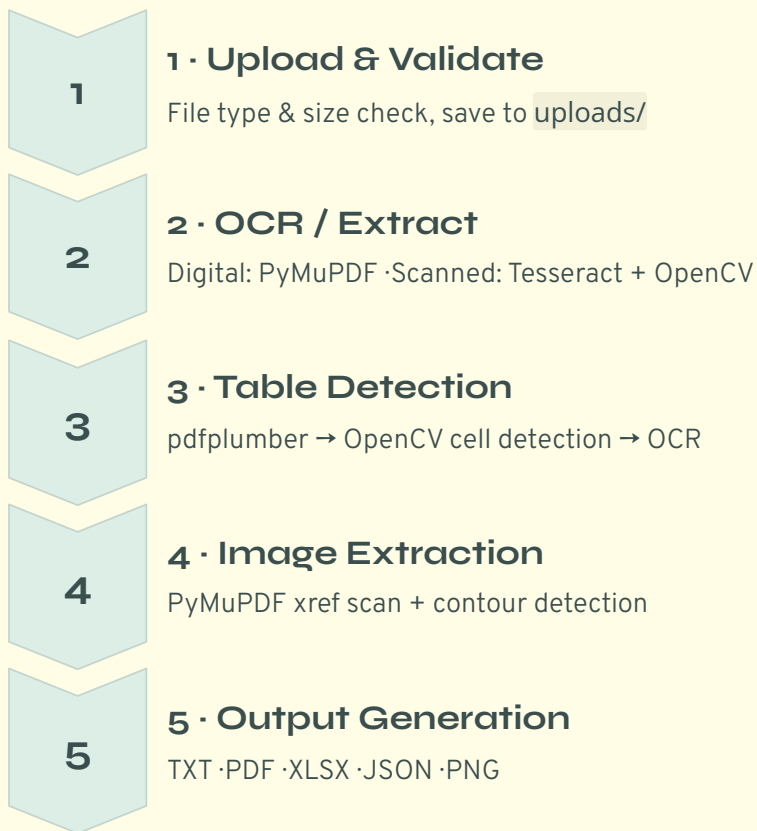
Receives files via REST, routes to the correct processor, returns results and download links.

Document Processor

Central orchestrator that fans out to OCR, table, image, and output sub-modules per page.

Processing Pipeline

Every uploaded file passes through the same fault-tolerant 5-stage pipeline – one failed page never stops the job.



📌 Fault-tolerant: Each page is processed independently – a single bad page is skipped with a warning, never crashing the entire job.

Key Features



Dual OCR Mode

Detects digital vs. scanned pages automatically.
PyMuPDF for digital, Tesseract for scanned.



Smart Preprocessing

OpenCV pipeline: deskew, denoise, adaptive threshold – maximizes OCR accuracy on skewed scans.



3-Tier Table Engine

pdfplumber → OpenCV grid detection → text heuristics.
Graceful fallback for any document type.



Logo & Image Extraction

PyMuPDF xref extraction for PDFs. OpenCV contour detection for standalone images and stamps.



5 Output Formats

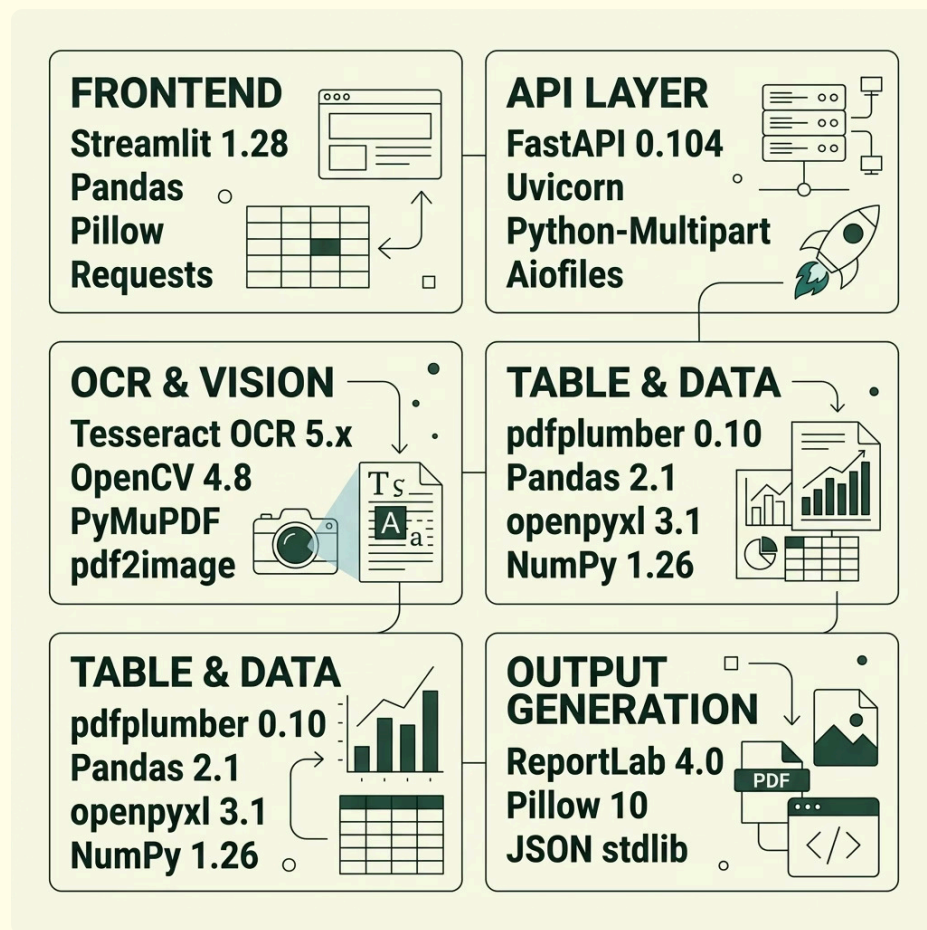
TXT, searchable PDF, Excel (multi-sheet), JSON (page-wise), PNG – all downloadable from the dashboard.



Fault Tolerant

Per-page error handling. Bad pages are skipped with a warning. The job never crashes on a single failure.

Technology Stack



Why This Stack?

Every library was chosen for production reliability and open-source maturity.

- **FastAPI** – async REST with auto-generated OpenAPI docs
- **Streamlit** – rapid, interactive UI with zero front-end boilerplate
- **Tesseract 5.x** – LSTM-based OCR, state-of-the-art accuracy
- **OpenCV 4.8** – battle-tested image preprocessing and contour detection
- **pdfplumber** – precise bounding-box table extraction for digital PDFs

Project Structure

```
idp_system/  
├── backend/  
│   ├── main.py  
│   ├── config.py  
│   └── processors/  
│       ├── ocr_engine.py  
│       ├── table_extractor.py  
│       ├── image_extractor.py  
│       └── output_generator.py  
├── utils/  
├── frontend/app.py  
├── outputs/ (6 sub-dirs)  
├── docker-compose.yml  
└── requirements.txt
```

main.py

FastAPI routes, upload handling, and download endpoints.

ocr_engine.py

Image preprocessing pipeline + Tesseract OCR integration.

table_extractor.py

3-tier table detection: pdfplumber → OpenCV → heuristics.

document_processor.py

Pipeline orchestrator with per-page fault tolerance.

frontend/app.py

Streamlit dashboard with all 5 extraction & download tabs.

Demo Results – Invoice

A real company invoice PDF was processed end-to-end – fields auto-extracted and ready for reuse in new invoices & e-Way Bills.

Apport Marine
SHIPPING COMPANY

INVOICE
Ref No. INV-2032-00124

Apport Marine Shipping Co.
123 Street Name City Name, State, Country, 12345
Tax ID: 00123456789012 | Tel: 00-123-4567
Email: info@apportmarine.com

Invoice Date: June 18, 2032
Due Date: June 25, 2032

Billed To:
Lorem Ipsum Co. Ltd.
123 Street Name City Name
State, Country: 12345
00-123-4567 or 00-123-4567(Fax)
loremipsum@example.com

Ship To:
Lorem Ipsum Co. Ltd.
123 Street Name City Name
State, Country: 12345
00-123-4567 or 00-123-4567(Fax)
loremipsum@example.com

No.	Description	Quantity	Unit Price (USD)	Amount (USD)
1	Ocean Freight – FCL 20' Container	1 container	\$1200.00	\$1200.00
2	Export Customs Clearance	1 shipment	\$90.00	\$90.00
3	Port Handling Fee – Origin	1 container	\$150.00	\$150.00
4	Bill of Lading Issuance	1 document	\$35.00	\$35.00
5	Documentation Handling	1 job	\$45.00	\$45.00
Subtotal				\$1800.00
Tax (VAT 7%)				\$126.00
Amount Due				\$1,926.00

Payment Info
Account Name: Apport Marine Shipping Co.
Account No.: 001-4-56789-0
Payment Due: June 25, 2032

Notes
Please send proof of payment to: invoices@apportmarine.com
For inquiries, contact our billing team at 00-123-4567

Michael B.
Authorized Signatures

Thank you for choosing our shipping services | Email: info@apportmarine.com | Website: www.apportmarine.com | Tel: 00-123-4567

- **Auto-fills new invoices** – Extracted data reused to generate future invoices in seconds
- **e-Way Bill generation** – Key fields like GSTIN, HSN, and amounts flow directly into e-Way Bills
- **Searchable records** – Extracted text is indexable and easy to retrieve
- **Audit-ready** – Structured data makes compliance checks faster
- **Scales easily** – Process hundreds of invoices at the same speed as one
- **Works on scanned docs** – OCR handles even non-digital or printed invoices
- **Multi-format output** – Data exported as Excel, JSON, or PDF for any workflow
- **Faster payments** – Quicker processing means fewer delays in the billing cycle

Demo Results – Invoice

A real company invoice PDF was processed end-to-end – fields auto-extracted and ready for reuse in new invoices & e-Way Bills.

Table → Excel Output

No	Description	Qty	Unit Price	Amount (USD)	Grand Total
1	Ocean Freight – FCL 20' Container	1 containe r	\$1,200.0 0	\$1,200.00	\$1,200.0 0
2	Export Customs Clearance	1 shipmen t	\$90.00	\$90.00	\$90.00
3	Port Handling Fee – Origin	1 containe r	\$150.00	\$150.00	\$150.00
4	Bill of Lading Issuance	1 documen t	\$35.00	\$35.00	\$35.00
Grand Total					\$1,926.0 0

Extracted Text Sample

Axport Marine Shipping Co.

Tax ID: 0123456789012

Invoice No: INV-2032-00124

Invoice Date: June 18, 2032 | Due: June 25, 2032

Bill To: Lorem Ipsum Co., Ltd.

Account No: 123-4-56789-0 | VAT 7%: \$126.00

Authorized Signatory: Michael B.

✔  Company logo + signature extracted → Vendor name, Invoice No., line items & amounts auto-fill new invoices & e-Way Bills in seconds

Conclusion & Future Scope

What We Built

- Full IDP pipeline in Python – PDF, JPG, JPEG, PNG
- Digital + scanned document handling via dual OCR
- 3-tier table extraction engine with graceful fallback
- OpenCV image preprocessing for skewed scans
- FastAPI REST backend + Streamlit dashboard
- Docker-ready deployment out of the box

Future Scope

- PaddleOCR for Hindi & regional script support
- LLM-powered Q&A over extracted documents
- Batch processing for multiple files at once
- Database storage for persistent job history
- Cloud deployment on AWS / GCP
- REST webhook notifications & mobile-responsive UI

Built with Python · FastAPI · Streamlit · OpenCV · Tesseract · PyMuPDF | Alliance University

